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Diverse Hedge Funds

Contents

1	Big picture	3
2	Data and measurement	4
2.1	Hedge fund universe and sample construction	4
2.2	Fund performance measures	4
2.3	Diversity as one minus network density	4
2.4	Five diversity dimensions	5
2.5	Control variables	5
3	Models and empirical specifications	6
3.1	Baseline multivariate performance regression	6
3.2	Fama-MacBeth regressions	6
3.3	Residual portfolio sorts	6
3.4	Event-study difference-in-differences	6
3.5	Instrumental-variable strategy	7
3.6	Solo-versus-team manager design	7
4	Identification and empirical design	8
4.1	Main identification challenge	8
4.2	Event-study design	8
4.3	Instrument relevance and exclusion	8
4.4	Manager-quality control through solo funds	8
4.5	Limitations	9
5	Tables and figures	10
5.1	Table 1: Summary statistics	10
5.2	Table 2 and Figure 1: Baseline performance regressions	11
5.3	Table 3: Portfolio sorts on team diversity	11
5.4	Table 4 and Figure 2: Event study	12
5.5	Table 5: Instrumental-variable analysis	12
5.6	Table 6: Managers who operate both solo- and team-managed funds	13
5.7	Table 7: Mechanisms - anomalies and behavioral biases	13
5.8	Table 8: Shareholder restrictions and capacity	13
5.9	Table 9: Investment risk, operational risk, and performance flags	14
5.10	Table 10: Capacity constraints and performance persistence	15
5.11	Table 11: Robustness tests	15
6	Short summary	16
7	Conclusion	16
7.1	Core conclusion	16
7.2	Economic intuition	16

7.3	Identification takeaway	16
7.4	Broader implication	16

1 Big picture

- **Question.** Do diverse hedge fund teams outperform homogeneous teams, and if so, through what mechanisms?
- **Main setting.** The paper studies hedge funds because they are sophisticated, unconstrained, typically team-managed, and less directly entangled with board-diversity channels than venture capital or private equity funds.
- **Main diversity dimensions.** The paper measures diversity in educational institution, college major, work experience, gender, and race.
- **Core empirical finding.** Hedge funds managed by diverse teams outperform those managed by homogeneous teams after adjusting for risk, fund characteristics, manager quality proxies, team size, strategy, and time effects.
- **Economic magnitude.** In baseline multivariate regressions, diverse teams outperform homogeneous teams by roughly 1.96% to 5.59% per year, depending on the diversity measure and return metric.
- **Performance breadth.** The positive diversity effect appears in raw returns, Fung-Hsieh alphas, residualized returns, Sharpe ratios, information ratios, manipulation-proof measures, and Berk-van Binsbergen value-added skill.
- **Endogeneity response.** The paper uses an event study of diversity-enhancing team transitions, an instrumental-variable design based on founder hometown racial diversity, and a within-manager design using managers who operate both solo- and team-managed funds.
- **Mechanism 1: broader opportunity set.** Diverse teams load on and arbitrage more stock market anomalies, suggesting they exploit a wider set of investment ideas.
- **Mechanism 2: fewer behavioral biases.** Diverse teams exhibit less disposition effect, overconfidence, and lottery-like trading behavior.
- **Mechanism 3: better risk management.** Diverse teams exhibit lower idiosyncratic risk, lower downside beta, lower maximum loss, and lower maximum drawdown.
- **Capacity and persistence.** Diverse teams appear less constrained by fund size and show more performance persistence, consistent with diversity expanding investment capacity.
- **Core takeaway.** Diversity is valuable in asset management because heterogeneous skills and perspectives improve idea generation, reduce behavioral errors, and improve risk control.

2 Data and measurement

2.1 Hedge fund universe and sample construction

- The paper studies hedge funds from commercial hedge fund databases and supplements them with manually collected manager background data.
- The sample focuses on team-managed hedge funds because diversity is defined across members of a fund management team.
- The paper excludes the first 18 months of returns to mitigate backfill bias, following concerns that funds often begin reporting only after good initial performance.
- Fund performance is observed at the monthly frequency, and most regression specifications include strategy fixed effects, year-month fixed effects, and team-size fixed effects.
- The final sample contains thousands of funds and tens of thousands of fund-month observations, with manager background information from educational, work, gender, and race sources.

Interpretation: The sample is designed to identify whether diversity among the actual portfolio managers operating a hedge fund predicts investment performance, not whether a hedge fund company as a whole is diverse.

2.2 Fund performance measures

The paper uses several performance measures.

- **RETURN:** monthly hedge fund return net of fees.
- **ALPHA:** monthly abnormal return from the Fung and Hsieh (2004) hedge fund factor model, with factor loadings estimated over the prior 24 months.
- **Residual performance:** residuals from regressions of fund performance on fund and team controls, with the constant term added back.
- **Other performance measures:** Sharpe ratio, information ratio, Goetzmann et al. manipulation-proof performance measure, and Berk-van Binsbergen value-added skill.

Interpretation: The use of multiple performance metrics is important because hedge fund returns can be nonlinear, stale, smoothed, and exposed to strategy-specific risk factors.

2.3 Diversity as one minus network density

For a team with N members, let C_i be the number of teammates who share a connection with manager i along a given dimension. For example, in the education measure, a connection exists if two managers attended the same university.

Team network density is

$$Density = \frac{1}{N} \sum_{i=1}^N \frac{C_i}{N-1}. \quad (1)$$

Team diversity is

$$Diversity = 1 - Density. \quad (2)$$

Interpretation: The measure equals zero when every member shares the same background and rises as within-team connections become sparse. It is naturally bounded and comparable across teams of different sizes.

2.4 Five diversity dimensions

- **DIVERSITY_EDU:** one minus network density based on shared educational institutions.
- **DIVERSITY_MAJOR:** one minus network density based on shared college majors.
- **DIVERSITY_EXP:** one minus network density based on shared work experience.
- **DIVERSITY_GENDER:** one minus network density based on gender.
- **DIVERSITY_RACE:** one minus network density based on race.

Interpretation: The first three dimensions are functional diversity measures: they proxy for differences in skills, training, and investment perspectives. Gender and race are demographic dimensions that also matter for communication, groupthink, and selection.

2.5 Control variables

The baseline controls include:

- team SAT score as a proxy for manager quality;
- management fees and performance fees;
- high-water-mark indicator;
- lockup period, redemption period, and fund age;
- leverage indicator;
- log fund size;
- fund strategy fixed effects;
- year-month fixed effects;
- team-size fixed effects.

Interpretation: The controls absorb standard determinants of hedge fund performance and contractual features that affect fund incentives, investor liquidity, and risk-taking.

3 Models and empirical specifications

3.1 Baseline multivariate performance regression

The baseline pooled OLS specification is

$$ALPHA_{i,m} = \alpha + \beta_1 Diversity_{i,m-1} + \beta_2 SAT_i/100 + \beta_3 MGT FEE_i + \beta_4 PER F FEE_i \quad (3)$$

$$+ \beta_5 HWM_i + \beta_6 LOCKUP_i + \beta_7 LEVERAGE_i + \beta_8 AGE_{i,m-1} \quad (4)$$

$$+ \beta_9 REDEMPTION_i + \beta_{10} \log(FUNDSIZE_{i,m-1}) \quad (5)$$

$$+ \sum_k \gamma_k YEARMTHDUM_{k,m} + \sum_l \delta_l STRATEGYDUM_{l,i} + \sum_o \phi_o TEAMSIZEDUM_{o,i} + \varepsilon_{i,m}. \quad (6)$$

The same specification is also estimated using fund return as the dependent variable.

Interpretation: The coefficient β_1 measures whether a fund with a more diverse team outperforms otherwise comparable funds in the same period, strategy, and team-size category.

3.2 Fama-MacBeth regressions

The paper also estimates monthly cross-sectional regressions and averages coefficients over time:

$$ALPHA_{i,m} = a_m + b_m Diversity_{i,m-1} + c'_m X_{i,m-1} + u_{i,m}, \quad (7)$$

where X includes the same controls as in the pooled specification.

Interpretation: Fama-MacBeth regressions reduce concerns that pooled OLS inference is driven by panel dependence and show whether diversity predicts performance in monthly cross-sections.

3.3 Residual portfolio sorts

The paper first removes the effects of fund and team controls by estimating the baseline performance regression without the diversity variable. It then sorts funds into diversity portfolios based on the residualized performance series.

Interpretation: This design asks whether performance differences remain after stripping out observable fund characteristics. It is a transparent complement to regression estimates.

3.4 Event-study difference-in-differences

The event study examines funds that add a new manager whose background increases team diversity. The treatment and control groups are formed as follows.

- Treatment funds add a manager from a different background relative to the existing team.
- Control funds start at the same diversity level and add a non-diversity-enhancing manager in the same event month.
- The event window runs from 36 months before to 36 months after the manager addition.
- Treatment funds are matched to control funds based on pre-event performance.

The difference-in-differences effect is

$$DD = (After_{Treat} - Before_{Treat}) - (After_{Control} - Before_{Control}). \quad (8)$$

Interpretation: This design controls for time-invariant fund differences and compares performance changes around manager additions that do versus do not increase diversity.

3.5 Instrumental-variable strategy

The instrument is founder hometown racial diversity. The logic is that founders who grew up in diverse hometowns may be more comfortable forming diverse teams.

The first stage is

$$Diversity_i = \pi_0 + \pi_1 HometownDiversity_i + \pi_2' X_i + \eta_i. \quad (9)$$

The second stage is

$$ALPHA_i = \alpha + \beta \widehat{Diversity}_i + \gamma' X_i + \varepsilon_i. \quad (10)$$

Interpretation: The IV design addresses unobserved time-varying differences between diverse and homogeneous funds, assuming founder hometown diversity affects fund performance only through team diversity.

3.6 Solo-versus-team manager design

The paper studies managers who simultaneously operate solo-managed and team-managed hedge funds. It relates diversity to the performance difference between team-managed funds and the average performance of the same managers' solo-managed funds.

$$PerformanceDiff_{i,m} = Performance_{i,m}^{Team} - AveragePerformance_{i,m}^{Solo}. \quad (11)$$

Interpretation: This design controls for manager quality by comparing each team to the solo funds operated by the same managers. The estimates likely understate diversity benefits because managers can transfer ideas from teams to their solo funds.

4 Identification and empirical design

4.1 Main identification challenge

The main challenge is that diverse funds may differ from homogeneous funds in unobservable ways.

- Better managers may choose diverse teams.
- Larger or more sophisticated firms may form diverse teams.
- Diversification of strategy, geography, or clientele may correlate with team diversity.
- Diverse funds may differ in investor restrictions, capacity, and risk management.

Interpretation: A positive diversity coefficient in a baseline regression is not automatically causal. The paper therefore layers several designs that target different endogeneity concerns.

4.2 Event-study design

The event study is useful because it focuses on changes within funds around manager additions.

- It compares funds before and after a diversity-enhancing team transition.
- It matches to funds that hire managers without increasing diversity.
- Figure 2 shows similar pre-trends for treatment and control funds.

Interpretation: The event study addresses time-invariant differences across funds, though it still relies on the assumption that diversity-enhancing hires are not made exactly when unobserved performance prospects improve.

4.3 Instrument relevance and exclusion

The paper argues that founder hometown racial diversity predicts team diversity through childhood imprinting.

- The first-stage relation is positive across all five diversity measures.
- First-stage F-statistics are close to or above common relevance thresholds.
- The authors conduct auxiliary tests linking hometown racial composition to team racial composition.

Interpretation: The IV estimates tend to be larger than OLS estimates, suggesting that OLS may understate the benefits of diversity or that the instrument identifies a particular high-value source of diversity.

4.4 Manager-quality control through solo funds

The solo-versus-team design controls for manager skill because the same managers appear in both solo and team settings.

- If diversity were merely a proxy for manager ability, the performance of the same managers' solo funds should absorb that effect.
- The paper finds that team-managed diverse funds still outperform relative to those managers' solo funds.

Interpretation: This design is one of the strongest checks because it compares diverse and homogeneous team performance after anchoring on the individual managers' own solo performance.

4.5 Limitations

- Diversity is not randomly assigned.
- Manager background information may be incomplete or noisy.
- The IV exclusion restriction cannot be directly tested.
- Hedge fund databases suffer from reporting selection and survivorship concerns, although the paper applies standard corrections.
- The mechanisms are inferred from holdings, trading behavior, and risk measures rather than direct observation of team discussions.

Interpretation: The empirical case is persuasive because many independent analyses point in the same direction, not because any one design is perfect.

5 Tables and figures

This section uses the original tables and figures extracted from the paper. Adjacent visuals are grouped compactly to save space while keeping the same *Read* plus *Economic message* format.

5.1 Table 1: Summary statistics

Read:

- Panel A lists common universities, majors, workplaces, genders, and races among hedge fund managers.
- Panel B reports that diversity in education, major, and work experience is relatively high, while gender and race diversity are lower.
- Panel C shows that diversity varies across investment strategies.
- Panel D reports positive correlations among diversity measures, especially among functional dimensions such as major and work experience.

Economic message: Hedge fund teams are diverse in some functional dimensions but much less diverse along demographic dimensions. This variation is the empirical basis for testing whether diversity predicts performance.

Table 1
Summary statistics

A: Universities, college majors, former workplaces, genders, and races of hedge fund managers

No.	University/Major/Workplace/Gender/Race	Number of managers	Percentage of managers
<i>1: Top ten universities</i>			
1	Harvard University	270	7.98
2	University of Pennsylvania	212	6.26
3	Columbia University	186	5.49
4	New York University	182	5.38
5	University of Chicago	115	3.40
6	Yale University	95	2.81
7	Cornell University	87	2.57
8	University of Virginia	78	2.30
9	Massachusetts Institute of Technology	73	2.16
10	Stanford University	71	2.10
<i>2: Top ten college majors</i>			
1	Finance	921	29.79
2	Economics	500	16.17
3	Accounting	204	6.60
4	Computer Science	172	5.56
5	Mathematics	168	5.43
6	History	97	3.14
7	Management	83	2.68
8	Physics	55	1.78
9	Commerce	43	1.39
10	Politics	35	1.13
<i>3: Top ten former workplaces</i>			
1	Goldman Sachs	153	4.52
2	Morgan Stanley	142	4.19
3	Merrill Lynch	129	3.81
4	JP Morgan	124	3.66
5	UBS	90	2.66
6	Credit Suisse	72	2.13
7	Deutsche Bank	68	2.01
8	Bear Stearns	61	1.80
9	Lehman Brothers	56	1.65
10	Citigroup	55	1.62
<i>4: Gender</i>			
1	Male	11829	94.12
2	Female	739	5.88
<i>5: Race</i>			
1	White	7319	64.83
2	Asian	1845	16.34
3	Black	1299	11.51
4	Hispanic	827	7.33

B: Distribution of diversity measures and key variables

Diversity measure/variable	Mean	25%	Median	75%	Std dev
DIVERSITY_EDU	0.789	0.100	1.000	1.000	0.393
DIVERSITY_MAJOR	0.742	0.476	1.000	1.000	0.416
DIVERSITY_EXP	0.560	0.000	1.000	1.000	0.490
DIVERSITY_GENDER	0.112	0.000	0.000	0.000	0.272
DIVERSITY_RACE	0.584	0.000	1.000	1.000	0.462
SAT	1434.680	1400.000	1475.000	1505.000	108.600
RETURN	0.449	-1.080	0.450	2.040	5.179
MGT FEE	1.426	1.000	1.500	2.000	0.588
PERF FEE	17.390	20.000	20.000	20.000	6.516
HWM	0.729	0.000	1.000	1.000	0.445
LOCKUP	0.256	0.000	0.000	0.250	0.517
LEVERAGE	0.592	0.000	1.000	1.000	0.492
AGE	6.468	2.583	5.083	8.917	5.244
REDEMPTION	2.063	1.000	1.000	3.000	2.656
FUND SIZE	441.380	18.900	68.540	249.960	2732.220

Table 1, panels A–B

Table 1
(Continued)

C: Distribution of diversity measures by investment strategy

Investment strategy	No. of funds	Mean	25%	Median	75%	Std dev
<i>1: Diversity in educational institution</i>						
Directional Trader	587	0.804	1.000	1.000	1.000	0.382
Relative Value	468	0.713	0.000	1.000	1.000	0.445
Security Selection	2152	0.790	1.000	1.000	1.000	0.394
Multiprocess	600	0.822	1.000	1.000	1.000	0.362
<i>2: Diversity in college major</i>						
Directional Trader	787	0.726	0.333	1.000	1.000	0.431
Relative Value	534	0.636	0.000	1.000	1.000	0.445
Security Selection	2457	0.759	0.533	1.000	1.000	0.407
Multiprocess	736	0.775	0.700	1.000	1.000	0.399
<i>3: Diversity in work experience</i>						
Directional Trader	649	0.597	0.000	1.000	1.000	0.486
Relative Value	426	0.472	0.000	0.125	1.000	0.490
Security Selection	2101	0.567	0.000	1.000	1.000	0.491
Multiprocess	631	0.554	0.000	1.000	1.000	0.488
<i>4: Diversity in gender</i>						
Directional Trader	2770	0.410	0.000	0.000	0.667	0.476
Relative Value	1151	0.293	0.000	0.000	0.667	0.432
Security Selection	6013	0.397	0.000	0.000	0.500	0.472
Multiprocess	1702	0.543	0.000	0.000	0.500	0.480
<i>5: Diversity in race</i>						
Directional Trader	2761	0.556	0.000	1.000	1.000	0.474
Relative Value	1149	0.508	0.000	0.553	1.000	0.454
Security Selection	5171	0.566	0.000	0.697	1.000	0.460
Multiprocess	1697	0.706	0.303	1.000	1.000	0.427

D: Correlations between diversity measures and key variables

Key variable	DIVERSITY_EDU	DIVERSITY_MAJOR	DIVERSITY_EXP	DIVERSITY_GENDER	DIVERSITY_RACE
DIVERSITY_EDU	1.000				
DIVERSITY_MAJOR	0.412	1.000			
DIVERSITY_EXP	0.547	0.692	1.000		
DIVERSITY_GENDER	0.066	0.066	0.050	1.000	
DIVERSITY_RACE	-0.061	0.021	0.016	0.199	1.000
SAT	0.649	0.264	0.472	0.243	0.183
RETURN	0.023	0.033	0.039	0.009	0.008
MGT FEE	0.038	0.022	0.000	0.056	-0.010
PERF FEE	-0.035	0.123	0.041	0.065	0.041
HWM	-0.076	0.000	-0.042	-0.024	0.050
LOCKUP	0.008	-0.056	-0.049	-0.055	-0.086
LEVERAGE	0.001	0.047	0.040	-0.036	-0.030
AGE	0.083	0.072	0.104	0.022	0.004
REDEMPTION	0.064	0.041	0.040	0.033	-0.033
FUND SIZE	-0.078	0.016	-0.018	-0.028	0.022
TEAM SIZE	-0.406	-0.159	-0.380	-0.013	0.057

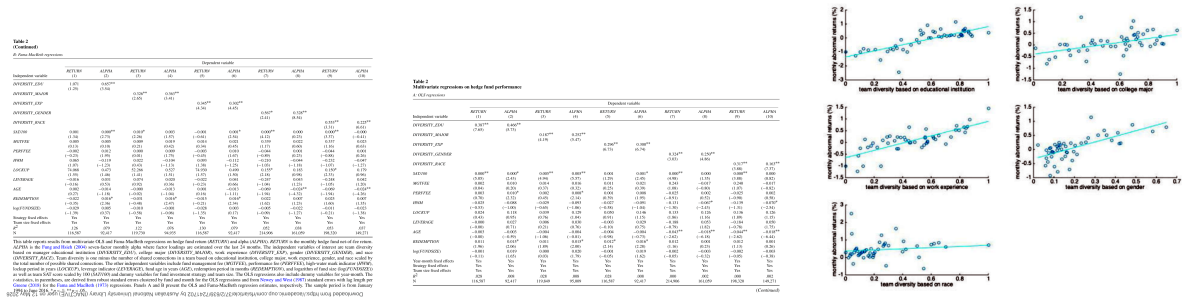
Table 1, panels C–D

5.2 Table 2 and Figure 1: Baseline performance regressions

Read:

- Table 2 shows positive and significant coefficients on most diversity measures in pooled OLS and Fama-MacBeth regressions.
- The dependent variables are monthly returns and Fung-Hsieh alphas.
- The economic effect ranges from roughly 1.96% to 5.59% per year depending on the diversity dimension and specification.
- Figure 1 provides binned scatter plots showing a positive relation between residualized fund abnormal returns and team diversity.

Economic message: The baseline evidence shows that diversity is not simply a proxy for strategy, team size, or fund contract terms. Diverse teams perform better after extensive controls.



5.3 Table 3: Portfolio sorts on team diversity

Read:

- Funds are sorted into portfolios based on diversity after residualizing performance with respect to fund and team controls.
- High-diversity portfolios outperform low-diversity portfolios across education, major, experience, gender, and race diversity.
- The high-minus-low alpha spread is economically meaningful and often statistically significant.

Economic message: Portfolio sorts provide a transparent nonparametric confirmation of the regression results: diversity is associated with superior risk-adjusted performance.

Table 3 Portfolio sorts on hedge fund team diversity													
Hedge fund portfolio	Number of funds	Residuals (annualized)	t-statistic of residuals	Alpha from residuals (annualized)	t-statistic of alpha	SNPMF	SCMLC	BIDRET	BAAMTSY	PTFSRD	PTFSFX	PTFSOM	Adj. R ²
A: Diversity in educational institution													
Portfolio 1 (high diversity)	2,562	5.04**	2.40	2.67*	2.08	0.27**	0.18**	-0.31	-2.39**	-0.00	0.02**	0.00	0.564
Portfolio 2	256	2.64	0.91	-0.36	-1.31	0.23**	0.11	0.46	-1.97	0.00	-0.00	0.04	0.838
Portfolio 3	301	3.36	2.84	-1.08	-1.08	0.17**	0.13**	-1.24**	-2.61**	-0.00	0.01	-0.00	0.481
Portfolio 4	268	0.86	0.77	-1.44*	-2.17	0.13**	0.10**	-1.11*	-3.14**	-0.01	0.00	0.00	0.274
Portfolio 5 (low diversity)	339	-0.36	-0.23	-2.52**	-3.20	0.24**	0.20**	-0.03	-1.41*	-0.01	0.00	-0.01	0.525
Spread (1-5)		5.40**	2.64	5.16**	4.91	0.05**	-0.02	-0.28**	-0.98**	0.01	0.02*	0.01	0.098
B: Diversity in college major													
Portfolio 1 (high diversity)	1,711	7.08**	3.42	4.20**	2.82	0.27**	0.18**	0.08	-2.40**	-0.01	0.02	-0.00	0.438
Portfolio 2	374	2.80	1.93	3.50**	3.72	0.34	0.16	6.45	-5.45	-0.07	0.00	0.07	0.541
Portfolio 3	360	4.08**	4.15	1.56*	2.12	0.25**	0.14**	-1.11**	-2.90**	-0.01**	0.01*	-0.00	0.742
Portfolio 4	365	0.80	0.45	0.84	0.61	0.27**	0.12**	-1.10*	-3.70**	-0.01*	0.00	-0.00	0.351
Portfolio 5 (low diversity)	331	-2.04	0.21	-1.80	-0.28	0.28**	0.17**	-0.34	-2.10**	-0.00	0.02**	0.00	0.566
Spread (1-5)		6.12*	2.41	6.06**	2.44	-0.01	0.00	6.42	-6.36	-0.01	0.00	0.00	0.091
C: Diversity in work experience													
Portfolio 1 (high diversity)	1,761	6.90**	2.91	2.16	1.64	0.28**	0.19**	-0.25	-2.24**	-0.00	0.02**	0.00	0.549
Portfolio 2	366	2.81	1.19	0.76	1.17	0.20**	0.10	-2.08	-6.54*	-0.00	-0.01	-0.01	0.441
Portfolio 3	442	4.20	0.98	1.44*	1.97	0.32**	0.20*	1.49	-5.58	-0.01*	0.02	0.02	0.323
Portfolio 4	399	-0.72	-0.10	-0.75	-0.90	0.36**	0.20	-2.13	-10.42**	0.04	-0.04	-0.01	0.325
Portfolio 5 (low diversity)	765	0.84	0.69	-2.28**	-2.86	0.23**	0.19**	-0.64	-1.73**	-0.01	0.01	-0.00	0.631
Spread (1-5)		6.12*	2.22	4.44**	3.85	0.05*	0.00	-0.39	-8.51	0.01	0.01**	0.00	0.182
D: Diversity in gender													
Portfolio 1 (high diversity)	672	6.72**	4.43	5.88**	7.05	0.37**	0.27**	-0.70	-2.51*	0.00	0.01*	0.02*	0.593
Portfolio 2	223	5.64	1.92	6.28**	2.20	0.21**	0.22	-1.26	-3.40**	-0.02	0.02	0.02	0.564
Portfolio 3	815	6.90**	3.01	3.60**	2.10	0.54**	0.55**	-2.14	-12.97**	0.01	0.02	0.02	0.623
Portfolio 4	832	4.62*	2.42	1.44	1.74	0.17**	0.17**	-0.87*	-3.90**	-0.01	0.01	-0.00	0.664
Portfolio 5 (low diversity)	9123	1.56	1.23	0.96	1.02	0.26**	0.15**	-0.97**	-2.95**	-0.00	0.01**	0.00	0.664
Spread (1-5)		5.16**	4.03	4.92**	7.59	0.11**	0.12**	0.27	-9.26	0.00	0.02	0.02	0.347
E: Diversity in race													
Portfolio 1 (high diversity)	5,872	5.88**	7.87	5.52**	4.17	0.27**	0.21**	-1.13*	-1.93**	-0.00	0.02**	0.01	0.204
Portfolio 2	655	4.02**	7.22	3.24**	1.14	0.31**	0.13**	-0.89**	-2.47**	-0.01	0.01	-0.00	0.830
Portfolio 3	781	4.32**	6.72	2.04*	2.33	0.31**	0.13**	-1.02**	-2.87**	-0.01*	0.01	-0.01	0.716
Portfolio 4	801	5.50**	3.62	0.96	1.00	0.25**	0.11**	-0.86**	-2.47**	-0.01	0.02**	0.00	0.826
Portfolio 5 (low diversity)	3,654	1.56*	2.89	0.55	0.65	0.27**	0.16**	-1.08**	-1.93**	-0.00	0.01**	0.01	0.643
Spread (1-5)		4.32**	4.57	4.97**	3.38	0.00	0.00	-0.13	-6.02	0.00	0.01	0.00	0.066

The table reports portfolio sorts that analyze the residuals from the regression of fund returns on the fund and team controls (from Equation 1) after adding back each team diversity factor. In January 1st, hedge funds are sorted into portfolios based on the diversity factor, which is defined as the number of possible shared characteristics. Portfolio performance is estimated relative to the Fung and Hsieh (2004) benchmark, which is S&P 500 return minus risk free rate (SNPMF). Russell 2000 return minus S&P 500 return (SCMLC), change in the spread of Moody's AAA bond over 10 year Treasury bond (BIDRET), change in the spread of Moody's AAA bond over 10 year Treasury bond (BAAMTSY), change in the spread of Moody's AAA bond over 10 year Treasury bond (PTFSRD), change in the spread of Moody's AAA bond over 10 year Treasury bond (PTFSFX), change in the spread of Moody's AAA bond over 10 year Treasury bond (PTFSOM), change in the spread of Moody's AAA bond over 10 year Treasury bond (PTFSRD), change in the spread of Moody's AAA bond over 10 year Treasury bond (PTFSFX), change in the spread of Moody's AAA bond over 10 year Treasury bond (PTFSOM), change in the spread of Moody's AAA bond over 10 year Treasury bond (PTFSRD), change in the spread of Moody's AAA bond over 10 year Treasury bond (PTFSFX), change in the spread of Moody's AAA bond over 10 year Treasury bond (PTFSOM), change in the spread of Moody's AAA bond over 10 year Treasury bond (PTFSRD), change in the 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The table reports portfolio sorts that analyze the residuals from the regression of fund returns on the fund and team controls from Equation (1) after adding back the constant term. Every January 1st, hedge funds are sorted into five portfolios based on team diversity, which is defined as one minus the number of shared connections in a team based on educational institution, college major, work experience, gender, and race scaled by the total number of possible shared connections. Portfolio performance is estimated relative to the Fung and Hsieh (2004) factors, which are S&P 500 return minus risk free rate (SNPMF), Russell 2000 return minus S&P 500 return (SCMLC), change in the constant maturity yield of the U.S. 10-year Treasury bond (BIDRET), change in the spread of Moody's BAA bond over 10-year Treasury bond (BAAMTSY), bond PTF (PTFSRD), currency PTF (PTFSFX), and commodity PTF (PTFSOM), where PTF is a primitive trend following strategy. Panels A, B, C, D, and E report results for team diversity based on educational institution, college major, work experience, gender, and race, respectively. The t-statistics are derived from White (1980) standard errors. The sample period is from January 1990 to December 2013.

5.4 Table 4 and Figure 2: Event study

Read:

- Table 4 compares funds that add diversity-enhancing managers with matched control funds that add non-diversity-enhancing managers.
- After diversity-enhancing hires, treatment funds improve more than controls in both returns and residualized performance.
- The post-minus-pre improvement is around 3% to 6% per year depending on the diversity dimension and performance metric.
- Figure 2 shows that treatment and control funds have similar pre-event trends, supporting the parallel-trends assumption.

Economic message: The event study supports a causal interpretation: performance improves after a fund becomes more diverse, relative to comparable funds that also hire managers but do not increase diversity.

Table 4
Event study with difference-in-differences analysis

Fund performance attribute	Fund performance				Fund residuals			
	Before (1)	After (2)	After - before (3)	t-statistic (4)	Before (5)	After (6)	After - before (7)	t-statistic (8)
<i>A: Diversity in educational institution</i>								
Fund return (percent/month), treatment group	0.547	0.890	0.343	2.00	0.447	0.657	0.210	1.76
Fund return (percent/month), control group	0.532	0.324	-0.208	-1.64	0.456	0.345	-0.111	-1.56
Difference in return (percent/month)			0.551*	2.58			0.321*	2.31
Fund alpha (percent/month), treatment group	0.288	0.694	0.406	1.71	0.203	0.435	0.232	1.98
Fund alpha (percent/month), control group	0.268	0.145	-0.123	-1.11	0.199	0.165	-0.034	-0.98
Difference in alpha (percent/month)			0.529*	2.02			0.266*	2.18
<i>B: Diversity in college major</i>								
Fund return (percent/month), treatment group	0.447	0.647	0.200	1.65	0.457	0.639	0.182	1.54
Fund return (percent/month), control group	0.445	0.237	-0.208	-2.21	0.489	0.378	-0.111	-1.67
Difference in return (percent/month)			0.408**	2.66			0.293*	2.16
Fund alpha (percent/month), treatment group	0.316	0.671	0.355	3.1	0.279	0.536	0.257	2.99
Fund alpha (percent/month), control group	0.318	0.172	-0.146	-1.98	0.251	0.034	-0.217	-1.45
Difference in alpha (percent/month)			0.501**	3.68			0.474**	2.75
<i>C: Diversity in work experience</i>								
Fund return (percent/month), treatment group	0.539	0.788	0.249	1.83	0.489	0.623	0.134	1.56
Fund return (percent/month), control group	0.532	0.245	-0.287	-2.11	0.452	0.273	-0.179	-1.56
Difference in return (percent/month)			0.536**	2.79			0.313*	2.18
Fund alpha (percent/month), treatment group	0.319	0.752	0.433	2.76	0.235	0.467	0.232	2.21
Fund alpha (percent/month), control group	0.309	0.213	-0.096	-0.91	0.278	0.204	-0.074	-1.99
Difference in alpha (percent/month)			0.529**	2.80			0.306**	2.75
<i>D: Diversity in gender</i>								
Fund return (percent/month), treatment group	0.439	0.656	0.267	1.61	0.476	0.698	0.222	1.98
Fund return (percent/month), control group	0.443	0.225	-0.218	-2.22	0.478	0.274	-0.204	-2.27
Difference in return (percent/month)			0.485*	2.52			0.425**	2.96
Fund alpha (percent/month), treatment group	0.226	0.607	0.381	3.11	0.223	0.439	0.216	2.11
Fund alpha (percent/month), control group	0.228	0.145	-0.083	-0.99	0.201	0.125	-0.076	-1.94
Difference in alpha (percent/month)			0.464**	3.13			0.292**	2.66
<i>E: Diversity in race</i>								
Fund return (percent/month), treatment group	0.497	0.657	0.160	1.68	0.467	0.595	0.128	1.87
Fund return (percent/month), control group	0.501	0.325	-0.176	-1.88	0.437	0.318	-0.119	-2.28
Difference in return (percent/month)			0.336*	2.52			0.247**	2.87
Fund alpha (percent/month), treatment group	0.332	0.587	0.255	1.69	0.267	0.438	0.171	1.99
Fund alpha (percent/month), control group	0.331	0.145	-0.186	-2.01	0.261	0.165	-0.096	-0.98
Difference in alpha (percent/month)			0.441*	2.49			0.267*	2.05

Table 4: event-study difference-in-differences

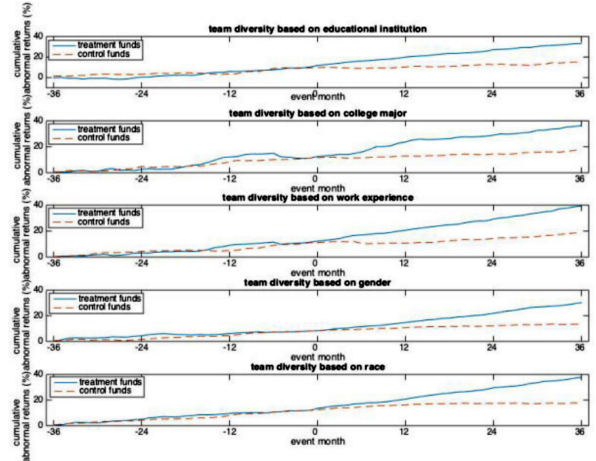


Figure 2: cumulative abnormal returns around diversity-enhancing hires

5.5 Table 5: Instrumental-variable analysis

Read:

- The first stage uses founder hometown racial diversity to predict team diversity.
- Hometown diversity is positively related to all five diversity measures.
- The second stage finds positive and significant effects of predicted diversity on fund alpha.
- The IV estimates are generally larger than the OLS estimates.

Economic message: The IV analysis suggests that the performance-diversity relation is not driven only by omitted fund characteristics. It also highlights childhood exposure to diversity as a source of team formation differences.

Table 7
Diversity, stock market anomalies, and behavioral biases

Independent variable				
DIVERSITY_EDU	DIVERSITY_MAJOR	DIVERSITY_EXP	DIVERSITY_GENDER	DIVERSITY_RACE
(1)	(2)	(3)	(4)	(5)
<i>A: Regressions on ANOMALY for all hedge funds</i>				
0.209**	0.155**	0.171**	0.101**	0.123**
(3.66)	(5.15)	(6.94)	(3.00)	(3.58)
<i>B: Regressions on ANOMALY for equity-focused hedge funds</i>				
0.312**	0.173**	0.179**	0.192**	0.120*
(4.83)	(4.05)	(4.15)	(3.35)	(2.31)
<i>C: Regressions on DISPOSITION for all hedge funds</i>				
-0.180**	-0.295**	-0.221**	-0.192**	-0.034**
(-3.25)	(-3.09)	(-6.35)	(-4.38)	(-2.73)
<i>D: Regressions on DISPOSITION for equity-focused hedge funds</i>				
-0.182**	-0.421**	-0.225**	-0.192**	-0.054**
(-3.34)	(-3.20)	(-6.34)	(-4.38)	(-3.60)
<i>E: Regressions on OVERCONFIDENCE for all hedge funds</i>				
-0.152**	-0.205**	-0.028**	-0.194*	-0.303*
(-4.56)	(-3.95)	(-2.63)	(-2.46)	(-2.49)
<i>F: Regressions on OVERCONFIDENCE for equity-focused hedge funds</i>				
-0.236**	-0.206**	-0.029**	-0.157**	-0.331**
(-3.51)	(-5.82)	(-3.19)	(-3.70)	(-3.29)
<i>G: Regressions on LOTTERY for all hedge funds</i>				
-0.013**	0.008**	-0.011**	-0.007**	-0.008**
(-4.53)	(-4.33)	(-6.05)	(-4.52)	(-4.65)
<i>H: Regressions on LOTTERY for equity-focused hedge funds</i>				
-0.013**	-0.014**	-0.011**	-0.006**	-0.009**
(-3.13)	(-4.86)	(-4.93)	(-3.26)	(-4.04)

Economic message: Diversity helps funds circumvent capacity constraints. With more stable capital, diverse teams can exploit a broader opportunity set without degrading performance.

5.9 Table 9: Investment risk, operational risk, and performance flags

Read:

- Panel A shows that diverse funds have lower idiosyncratic risk, lower downside beta, lower maximum loss, and lower maximum drawdown.
- Panel B shows that diverse funds have lower termination risk and fewer operational risk indicators.
- Panel C shows that diverse funds are less likely to exhibit suspicious performance flags such as repeated returns, kinks, high R-squared, or a high fraction of negative months.

Economic message: Diversity is not associated with reckless risk-taking. Instead, diverse teams deliver stronger performance with lower downside and operational risk.

Table 9
Multivariate regressions on hedge fund investment risk, operational risk, and performance flags

Independent variable				
DIVERSITY_EDU	DIVERSITY_MAJOR	DIVERSITY_EXP	DIVERSITY_GENDER	DIVERSITY_RACE
(1)	(2)	(3)	(4)	(5)
<i>1: Regressions on IDIORISK</i>				
-2.590**	-0.828**	-1.567**	-0.656**	-0.351**
(-3.93)	(-4.55)	(-6.04)	(-4.65)	(-4.03)
<i>2: Regressions on DOWNSIDEBETA</i>				
-0.192*	-0.184**	-0.211**	-0.181**	-0.113**
(-2.23)	(-3.09)	(-5.10)	(-3.59)	(-3.01)
<i>3: Regressions on MAXLOSS</i>				
-1.965*	-1.588**	-1.040**	-1.372**	-0.830**
(-2.25)	(-4.09)	(-3.59)	(-4.13)	(-3.72)
<i>4: Regressions on MAXDRAWDOWN</i>				
-3.967**	-4.213**	-1.467**	-2.016**	-0.914**
(-2.96)	(-6.86)	(-2.95)	(-3.61)	(-2.78)
<i>B: Regressions on fund operational risk</i>				
DIVERSITY_EDU	DIVERSITY_MAJOR	DIVERSITY_EXP	DIVERSITY_GENDER	DIVERSITY_RACE
(1)	(2)	(3)	(4)	(5)
<i>1: Logit regressions on TERMINATION</i>				
-0.563**	-0.146*	-0.230**	-0.231**	-0.359**
(-5.93)	(-2.25)	(-3.77)	(-4.16)	(-8.28)
[0.006]	[0.002]	[0.002]	[0.002]	[0.002]
<i>2: Cox regressions on TERMINATION</i>				
-0.526**	-0.139*	-0.223**	-0.206**	-0.265**
(-5.47)	(-2.26)	(-3.77)	(-3.90)	(-5.92)
<i>3: Logit regressions on VIOLATION</i>				
-1.610**	-1.066**	-0.561**	-0.679*	-0.283*
(-4.86)	(-6.09)	(-3.93)	(-2.17)	(-2.22)
[0.380]	[0.171]	[0.132]	[0.151]	[0.058]
<i>4: OLS regressions on OMEGA</i>				
-0.177**	-0.264*	-0.170**	-0.125**	-0.142**

Table 9, panels A–B

Table 8
Diversity and fund shareholder restrictions

Regressions on ALPHA														
Independent variable					DIVERSITY_EXP									
DIVERSITY_EDU					DIVERSITY_MAJOR					DIVERSITY_GENDER				
Low	Middle	High	Low	Middle	High	Low	Middle	High	Low	Middle	High	Low	Middle	High
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
<i>A: Funds sorted on redemption period</i>														
0.057	0.325*	0.462**	-0.007	0.162*	0.352**	0.183*	0.278**	0.371**	0.178	0.268**	0.403**	0.153*	0.203**	0.299**
(0.41)	(2.02)	(4.35)	(-0.11)	(2.26)	(4.45)	(2.50)	(3.35)	(4.22)	(1.83)	(3.84)	(6.75)	(2.44)	(5.38)	(4.53)
<i>B: Funds sorted on notice period</i>														
0.188	0.441**	0.549**	-0.254	-0.171	0.257**	-0.160	0.265**	0.475**	0.218**	0.279**	0.334**	0.346**	0.231**	0.207**
(1.49)	(3.04)	(4.28)	(-1.52)	(-1.81)	(3.80)	(-1.48)	(3.02)	(9.96)	(2.63)	(2.95)	(8.13)	(3.86)	(5.40)	(4.61)
<i>C: Funds sorted on lockup period</i>														
0.321**	0.583**	0.746**	-0.080	0.401	0.572**	0.206**	0.450**	0.711**	0.236**	0.115	0.516**	0.211**	0.215	0.276**
(2.65)	(2.91)	(3.22)	(-1.79)	(1.62)	(6.00)	(3.16)	(4.06)	(7.58)	(4.89)	(0.54)	(6.50)	(8.81)	(1.64)	(3.65)

This table reports results from multivariate OLS regressions on fund alpha for funds that are first sorted on their shareholder restrictions. The dependent variable is Fung and Hsieh (2004) seven-factor monthly fund alpha where factor loadings are estimated over the last 24 months (ALPHA). The independent variables of interest are team diversity based on manager educational institution (DIVERSITY_EDU), college major (DIVERSITY_MAJOR), work experience (DIVERSITY_EXP), gender (DIVERSITY_GENDER), and race (DIVERSITY_RACE). Team diversity is one minus the number of shared connections in a team based on educational institution, college major, work experience, gender, and race scaled by the total number of possible shared connections. The other independent variables include fund characteristics, such as the management fee (MGTFEE), performance fee (PERFEE), high-water mark indicator (HWM), leverage indicator (LEVERAGE), fund age in years (AGE), and logarithm of fund size (log(FUND SIZE)) as well as team SAT score scaled by 100 (SAT100) and dummy variables for year-month, fund investment strategy, and team size. The coefficient estimates on the fund and team control variables are omitted for brevity. The t-statistics, in parentheses, are derived from robust standard errors clustered by fund and month. The low, middle, and high redemption period groups in panel A comprise funds with redemption periods that do not exceed 15 days, that exceed 15 days but do not exceed one month, and that exceed one month, respectively. The low, middle, and high notice period groups in panel B are defined analogously. The low, middle, and high lockup period groups in panel C comprise funds with no lockups, with lockup periods that are less than or equal to a year, and with lockup periods that exceed a year, respectively. The sample period is from January 1994 to June 2016. * $p < 0.1$, ** $p < 0.05$.

Table 9
(Continued)

Independent variable				
DIVERSITY_EDU	DIVERSITY_MAJOR	DIVERSITY_EXP	DIVERSITY_GENDER	DIVERSITY_RACE
(1)	(2)	(3)	(4)	(5)
<i>C: Regressions on fund performance flags</i>				
<i>1: Regressions on %NEGATIVE</i>				
-0.220	-0.914**	-0.214**	-0.104	-0.306**
(-1.66)	(-8.33)	(-2.78)	(-1.47)	(-5.90)
[0.034]	[0.160]	[0.036]	[0.016]	[0.046]
<i>2: Regressions on KINK</i>				
-0.474**	-0.351*	-0.525**	-0.369**	-0.112**
(-3.44)	(-4.07)	(-6.99)	(-5.97)	(-2.91)
[0.108]	[0.086]	[0.102]	[0.084]	[0.031]
<i>3: Regressions on MAXRSQ</i>				
-1.092**	-1.408**	-0.436**	-0.944**	-0.122*
(-6.97)	(-9.16)	(-5.60)	(-6.45)	(-2.52)
[0.066]	[0.084]	[0.148]	[0.446]	[0.012]
<i>4: Regressions on %REPEAT</i>				
-0.498**	-0.448**	-0.500**	-0.275**	-0.016
(-3.90)	(-5.30)	(-7.19)	(-4.56)	(-0.41)
[0.135]	[0.113]	[0.115]	[0.064]	[0.005]

Table 9, panel C

5.10 Table 10: Capacity constraints and performance persistence

Read:

- Panel A studies how fund size affects performance across high-, medium-, and low-diversity groups.
- The negative size-performance relation is weaker or reversed for high-diversity funds.
- Panel B double sorts funds by diversity and past performance.
- Performance persistence is concentrated in high-diversity funds.

Economic message: Diverse teams are better able to scale and maintain performance, suggesting that heterogeneous perspectives expand capacity and make skill more persistent.

Table 10
Diversity, fund capacity constraints, and fund performance persistence

A: Regressions on fund performance for funds sorted by team diversity

Independent variable	Diversity in educational inst.				Diversity in college major				Diversity in work experience				Diversity in gender				Diversity in race			
	High (1)	Medium (2)	Low (3)		High (4)	Medium (5)	Low (6)		High (7)	Medium (8)	Low (9)		High (10)	Medium (11)	Low (12)		High (13)	Medium (14)	Low (15)	
1. Regressions on RETURN log(FUND02Z)	0.060** (3.21)	0.005 (0.30)	-0.056** (-4.04)		0.190** (8.96)	0.083** (3.63)	-0.026 (-1.09)		0.061* (2.48)	-0.026 (-1.32)	-0.047** (-2.59)		0.046** (3.28)	0.009 (0.84)	-0.064** (-5.67)		0.106** (7.11)	0.002 (0.18)	-0.065** (-4.41)	
2. Regressions on ALPHA log(FUND02Z)	0.067* (2.18)	0.056* (2.22)	-0.018** (-2.54)		0.089** (3.90)	0.000 (0.30)	-0.079** (-4.50)		0.083** (3.48)	-0.009 (-0.41)	-0.027 (-1.64)		0.068** (3.22)	-0.018 (-1.05)	-0.054** (-5.83)		0.072** (8.28)	-0.016 (-1.01)	-0.076** (-3.81)	

B: Double sorts on team diversity and past fund performance

Hedge fund portfolio	Diversity in educational inst.			Diversity in college major			Diversity in work experience			Diversity in gender			Diversity in race		
	High (1)	Medium (2)	Low (3)	High (4)	Medium (5)	Low (6)	High (7)	Medium (8)	Low (9)	High (10)	Medium (11)	Low (12)	High (13)	Medium (14)	Low (15)
1. Double sort on team diversity and past 24-month fund return															
Portfolio 1 (high past 24m alpha)	6.37	6.09	-0.67												
Portfolio 2	2.20	1.78	0.14	7.21	2.04	-1.44	7.34	1.76	1.44	5.78	0.89	1.66	6.11	5.47	3.22
Portfolio 3	1.54	0.56	-1.66	2.01	-0.67	1.86	2.22	0.76	-0.21	3.09	2.54	3.99	4.89	2.89	2.11
Portfolio 4	3.89	2.34	-2.45	3.87	0.38	-0.91	4.39	-2.21	-1.11	2.90	2.78	2.21	2.21	2.45	1.98
Portfolio 5 (low past 24m alpha)	6.58	-2.22	1.45	1.21	-0.81	0.34	1.12	0.58	1.34	-1.09	2.01	2.23	-1.43	1.90	1.49
Spread (1-5)	6.21**	3.21*	-2.12	6.00**	2.85*	-1.78	6.22**	1.20	0.10	6.87**	-1.12	-0.67	7.54**	3.57**	1.42
2. Double sort on team diversity and past 24-month fund return															
Portfolio 1 (high past 24m return)	6.34	3.14	-0.34												
Portfolio 2	3.45	2.78	1.66	8.72	3.44	4.21	6.78	3.12	1.09	5.78	5.04	2.88	7.32	4.32	3.21
Portfolio 3	2.89	1.56	-0.56	4.23	0.89	4.55	4.21	2.56	0.89	2.21	4.52	6.01	5.21	4.10	3.96
Portfolio 4	1.56	0.56	-2.56	3.23	0.78	-2.21	3.09	-1.01	-2.01	-1.09	3.12	3.45	3.12	2.98	2.87
Portfolio 5 (low past 24m return)	-0.44	1.12	0.26	2.36	2.21	1.99	1.44	2.01	0.45	2.88	1.98	1.99			
Spread (1-5)	6.78**	2.22*	-0.60	6.90**	1.13	0.22	5.34**	3.11*	-1.12	4.74**	2.85*	2.43	4.44**	2.36*	1.22

Panel A reports results from multivariate regressions on hedge fund performance for funds sorted by fund management team diversity. Team diversity is one minus the number of shared connections in a team based on educational institution, college major, work experience, gender, and race scaled by the total number of possible shared connections. The dependent variables include hedge fund return (RETURN) and alpha (ALPHA). RETURN is the monthly hedge fund net-of-fee return. ALPHA is the long and short (LS) seven-factor monthly alpha where factor loadings are estimated over the last 24 months. The management variable of interest is the logarithm of fund size (log(FUND02Z)). The other independent variables include fund characteristics, such as the management fee (MGTFEE), performance fee (PERFEE), high-water mark indicator (HWM), hedge fund age in years (AGE), average returns (AVERAGE), fund age in years (AGE), and management group in months (MANAGEMENT). Panel B reports double sorts on team diversity and past fund performance. Every January 1st, hedge funds are sorted into three groups based on team diversity in educational institution, college major, work experience, gender, or race. Thereafter, hedge funds in each group are sorted into the group based on past 24-month fund return and hedge (24m) alpha (past 24m return) or on past 24-month fund return (R2). The t-ratios are derived from White (1980) standard errors. The sample period is from January 1994 to June 2016. *p < .1, **p < .05.

Table 11
Robustness tests

Independent variable	Regressions on RETURN										Regressions on ALPHA									
	INTEGRITY ZSCORE (1)	INTEGRITY ZSCORE (2)	INTEGRITY ZSCORE (3)	INTEGRITY ZSCORE (4)	INTEGRITY ZSCORE (5)	INTEGRITY ZSCORE (6)	INTEGRITY ZSCORE (7)	INTEGRITY ZSCORE (8)	INTEGRITY ZSCORE (9)	INTEGRITY ZSCORE (10)	INTEGRITY ZSCORE (11)	INTEGRITY ZSCORE (12)	INTEGRITY ZSCORE (13)	INTEGRITY ZSCORE (14)	INTEGRITY ZSCORE (15)	INTEGRITY ZSCORE (16)	INTEGRITY ZSCORE (17)	INTEGRITY ZSCORE (18)	INTEGRITY ZSCORE (19)	
A. Herfindahl-Hirschman index (diversity measure)	0.628 (1.87)	0.212* (2.17)	0.504** (4.90)	0.549** (7.02)	0.315** (8.70)	0.448** (2.72)	0.377** (2.18)	0.448** (8.70)	0.448** (8.70)	0.448** (8.70)	0.448** (8.70)	0.448** (8.70)	0.448** (8.70)	0.448** (8.70)	0.448** (8.70)	0.448** (8.70)	0.448** (8.70)	0.448** (8.70)	0.448** (8.70)	
B. Double sorts on team diversity and past fund performance																				
Portfolio 1 (high past 24m alpha)	6.37	6.09	-0.67																	
Portfolio 2	2.20	1.78	0.14	7.21	2.04	-1.44	7.34	1.76	1.44	5.78	0.89	1.66	6.11	5.47	3.22					
Portfolio 3	1.54	0.56	-1.66	2.01	-0.67	1.86	2.22	0.76	-0.21	3.09	2.54	3.99	4.89	2.89	2.11					
Portfolio 4	3.89	2.34	-2.45	3.87	0.38	-0.91	4.39	-2.21	-1.11	2.90	2.78	2.21	2.21	2.45	1.98					
Portfolio 5 (low past 24m alpha)	6.58	-2.22	1.45	1.21	-0.81	0.34	1.12	0.58	1.34	-1.09	2.01	2.23	-1.43	1.90	1.49					
Spread (1-5)	6.21**	3.21*	-2.12	6.00**	2.85*	-1.78	6.22**	1.20	0.10	6.87**	-1.12	-0.67	7.54**	3.57**	1.42					

The table reports results from multivariate OLS regressions on hedge fund return (RETURN) and alpha (ALPHA). RETURN is the monthly hedge fund net-of-fee return. ALPHA is the long and short (LS) seven-factor monthly alpha where factor loadings are estimated over the last 24 months. The independent variables of interest are team diversity based on management educational institution (INTEGRITY_ZSCORE), college major (INTEGRITY_ZSCORE), work experience (INTEGRITY_ZSCORE), gender (INTEGRITY_ZSCORE), and race (INTEGRITY_ZSCORE). Team diversity is one minus the number of shared connections in a team based on educational institution, college major, work experience, gender, and race scaled by the total number of possible shared connections. The other independent variables include fund characteristics, such as the management fee (MGTFEE), performance fee (PERFEE), high-water mark indicator (HWM), hedge fund age in years (AGE), average returns (AVERAGE), fund age in years (AGE), and management group in months (MANAGEMENT). Panel B reports double sorts on team diversity and past fund performance. Every January 1st, hedge funds are sorted into three groups based on team diversity in educational institution, college major, work experience, gender, or race. Thereafter, hedge funds in each group are sorted into the group based on past 24-month fund return and hedge (24m) alpha (past 24m return) or on past 24-month fund return (R2). The t-ratios are derived from White (1980) standard errors. The sample period is from January 1994 to June 2016. *p < .1, **p < .05.

5.11 Table 11: Robustness tests

Read:

- The robustness tests use alternative diversity measures such as Herfindahl-Hirschman-based diversity.
- They include all five diversity measures jointly.
- They consider different alpha estimation windows and alternative sample restrictions.
- The positive diversity-performance relation remains robust.

Economic message: The findings do not depend on a single diversity definition, specification, or sample choice.

6 Short summary

- The paper studies whether diversity adds value in hedge fund management teams.
- Diversity is measured using educational institution, college major, work experience, gender, and race.
- Diversity is defined as one minus team network density based on shared connections.
- Baseline regressions show that diverse hedge funds outperform homogeneous funds.
- Portfolio sorts confirm that high-diversity funds deliver higher residualized performance.
- Event studies show that funds improve after diversity-enhancing manager additions.
- IV regressions using founder hometown racial diversity support a causal interpretation.
- Within-manager comparisons show that diverse teams outperform even after controlling for manager quality using solo-managed funds.
- Diverse teams exploit more stock anomalies and avoid behavioral biases.
- Diverse teams have lower downside risk, operational risk, and suspicious performance flags.
- Diversity helps funds overcome capacity constraints and generate persistent performance.
- The evidence suggests that diversity creates value through heterogeneous skills, better monitoring, and broader investment opportunity sets.

7 Conclusion

7.1 Core conclusion

Lu, Naik, and Teo show that hedge fund teams with more heterogeneous backgrounds outperform homogeneous teams. The result is not limited to one diversity dimension or one return metric; it appears across education, major, work experience, gender, and race diversity and across several performance measures.

7.2 Economic intuition

Diverse teams combine different training, networks, and perspectives. This allows them to identify more investment opportunities, arbitrage more stock anomalies, avoid groupthink and behavioral biases, and manage downside and operational risk more effectively. The same heterogeneity also helps funds scale without exhausting their best ideas.

7.3 Identification takeaway

The paper’s strongest claim comes from the convergence of multiple empirical strategies. Baseline regressions show a robust association, event studies show improvement after diversity rises, IV estimates use founder hometown diversity as exogenous variation, and within-manager tests compare team performance with the same managers’ solo funds.

7.4 Broader implication

The paper provides evidence that diversity has direct investment value in sophisticated asset management. It also suggests that allocator attention to diversity may be financially relevant, not merely a social or governance preference.